

DATABASE MANAGEMENT SYSTEMS (CS301)

Database Architecture Overview

Learning Objectives

Understand database architecture, three-level architecture, data independence, and client-server database models.

Introduction to Database Architecture

Database architecture defines how database systems are designed and organized. It specifies how users interact with data and how data is stored internally.

Three-Level Architecture

The ANSI/SPARC three-level architecture separates the database into three different levels to provide abstraction and data independence.

External Level

The external level is the user view level. Different users can have different views of the same database based on their requirements.

Conceptual Level

The conceptual level describes the overall structure of the entire database including entities, relationships, and constraints.

Internal Level

The internal level describes how data is physically stored in storage devices, including file organization and indexing methods.

Data Independence

Physical Data Independence: Ability to modify physical storage without affecting the conceptual schema.

Logical Data Independence: Ability to modify conceptual schema without affecting user views.

Two-Tier Architecture

In a two-tier architecture, the client communicates directly with the database server. It is commonly used in small-scale applications.

Three-Tier Architecture

In a three-tier architecture, an application server acts as an intermediary between the client and database server, improving scalability and security.

Client-Server Architecture

Client-server architecture divides work between clients that request services and servers that provide database services.

Advantages of Three-Level Architecture

- Improved security
- Better data abstraction
- Easier maintenance
- Supports data independence

Applications

- Banking Systems
- E-Commerce Applications
- Enterprise Resource Planning (ERP)
- University Management Systems

Exam Important Questions

1. Explain three-level database architecture.
2. Define data independence.
3. Differentiate logical and physical data independence.
4. Explain two-tier architecture.
5. Explain three-tier architecture.

Summary

Database architecture provides a structured framework for organizing databases. The three-level architecture and data independence concepts improve flexibility, security, and maintainability.